RVSP

UNIT-I: The Random Variable:

Introduction, Review of probability theory, Defenition of a Random variable, conditions for a function to be a Random Variable Discrete, continuous and Mixed Random Variables, Distribution and Density functions, properties, Bionomial, Poisson, Uniform, Gaussian, exponential, Rayleigh, Conditional Distribution, Conditional Density, properties.

UNIT-II: Operation on one Random Variable-Expectations:

Introduction, expected value of a Random Variable, function of a Random Variable, Moments about the Origin, Central Moments, variance and skew, chebychev's Inequality, characteristic Function, Moment Generating Function, Transformations of a Random Variable: Monotonic Trans formations for a continuos Random Variable, Non-monotonic Transformations of continuous Random Variable.

UNIT-III: Multiple Random Variables:

Vector Random variables, joint Distribution function, properties of joint Distribution, marginal Distribution functions, conditional D.F, density, statistical independence, sum of two random variables, sum of several Random variables, Central limit theorem: unequal Distribution, Equal distribution.

Operations on multiple Random Variables: joint moments that about the Origin, joint central moments, joint char. functions, jointly Gaussian R.V: Two R.V case, N R.V case, Properties, Transformations of multiple R.V, Linear Transformations of Gaussian R.V.

UNIT-IV: Random processes - Temporal characteristics:

The Random process concept, classification of processes, Deterministic & Nondeterministic processes, Distribution and density functions, concept of stationarity, Nth-order & strict-sense stationarity, Time Averages, Second-order & ergodicity, Autocorrelation function & its properties, cross-correlation function & its properties, Covariance functions, Gaussian R.P, poisson R.P.

UNIT-V: Random processes-spectral characteristics:

The power density spectrum properties, Relationship b/w power density spectrum & autocorrelation function, the cross-power D.S, prop, Relation b/w cross-power D.S and cross-Correlation function.

27/09/22

RVSP

Random Variables & stochartic processes

Communication: Transfer of data from source to destination.

Signal: It is function of time having one (or) more indepen dent vairables (conveys data transfer).

It is denoted by $x(t)$. where, x is dependent variable, t is independent variable

Signal formats: $x(t)$ – voice (one variable) (1D)
$f(x,y)$ – Image (2 variables 2D)
$f(x,y,t)$ – vedeo. (3 variables 3D)

Importance of signal: To convey Information.

* To do the communication, input is needed that is "signal".
* To do the communication in ~~a~~ long distance we need "modulation".
* Carrier – High frequency signal.
* Modulation: Mixing of message signal with carrier and transmitted via Anteena process.
* stochartic process – Random process ($x(t)$).

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UNIT-I

RANDOM VARIABLE

Experiment: It is the systematic procedure to collect all component and connect in the form of circuit, giving supply voltage and finally to note down the result.

There are 2 types of experiments – 1. Random exp.
2. Determnistic exp.

1. Random Exp: The exp. whose output cannot be predicted before the process is called Random exp.
   ex: Tossing a coin - Outcome cannot be predicted.

2. Determnistic exp: The exp. whose Output can be predicted due to some causes that is called determnistic exp.
   Ex: Throwing a stone is an exp - Output can be predicted i.e, it reaches on the earth due to the gravitational force.

Samplespace (S): Collection of all elements in a given exp. is called sample space, & denoted by S.

Ex: Tossing a coin.

$S = \{H, T\} = 2$

2. Tossing a 2 coins. $S = 2^2 = 4 = \{HH, HT, TH, TT\} = 4$

3. Tossing a 3 coins $S = 2^3 = 8 =$

4. Rolling a die $S = \{1, 2, 3, 4, 5, 6\} = 6$.

0	0	0	= TTT
0	0	1	= TTH
0	1	0	= THT
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

sample space.

Event: Experiment element (or) result of an exp.
(E)

Probability: $P(E) = \frac{F}{S} = \frac{\text{favorable no. of event}}{\text{Total no. of event}}$

favourable event: In a given exp. whatever assume the event i.e getting in the output that event is called favourable event.

Probability: It denotes the chance (It may occur (or) it may not Occur). It is the mathematical concept to measure degree of certainity (or) uncertainity, occurance & non occurance of an event.

Note:- The possibility of event which are getting in the output is called "exhaustic event".

- The event which is not possible getting an Output for the possible exp. Ex: Rolling a die (0, 7, 8 - -).

Prob

Ex: Rolling a die is a experiment, find prob of getting 6.

Experiment is Rolling a die,

$$S = \{1,2,3,4,5,6\} = 6.$$

$$F = \{6\} = 1.$$

$$P(E) = \frac{1}{6} \Rightarrow P(E) = \frac{1}{6}.$$

find probability of not getting event 6.

$$P(\bar{E}) = 1 - P(E) = 1 - \frac{1}{6} = \frac{5}{6}.$$

$$P(E) + P(\bar{E}) = 1.$$

Types of the probability

1. Individual probability
2. Joint probability
3. conditional probability
4. Total probability.

Probability (or) Individual probability: The Occurance of event and its chance is denoted by individual probability.

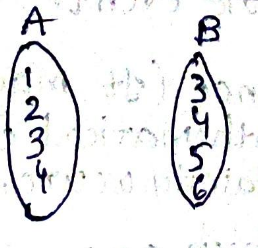
$$\boxed{P(A) = \frac{F}{S}}$$ A = occurance of event.

Joint probability: The Occurance of two events simultaneously and their chance is denoted by joint probability.

It is a mathematically represented by

$$\boxed{P(A \cap B) = P(A) + P(B) - P(A \cup B)}$$

Ex:



$A \cap B = \{3, 4\}$

$A \cup B = \{1, 2, 3, 4, 5, 6\}$

Conditional probability: The Occurance of which is the happened is denoted by conditional probability.

It is mathematically represented by

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

'A' denotes the Occurance of event
'B' denotes the happened event

Total probability: The occurance of one event from group of cells. its probability is denoted by total probability

$$\boxed{P(A) = \sum_{i=1}^{N} P(B_N) P(A/B_N)}$$

$$= P(B_1) P(A/B_1) + P(B_2) P(A/B_2) + P(B_3) P(A/B_3) + \ldots$$

Problem

In a box there are 100 resistors and tolerance values given in the table.

Resistor (Ω)	Tolerance 5%	10%	Total
22	10	14	24
47	28	16	44
100	24	8	32
Total	62	38	100

Let a resistor to be selected from a box and assume that each resistor has same likelihood choosen.

Define three events.

Event A: Draw a 47Ω Resistor
Event B: Draw resistor with 5% tolerance
Event C: Draw 100Ω Resistor
find Individual, joint, conditional probability.

Individual probability

$$P(A) = \frac{F}{S} = \frac{44}{100}$$

P(A) denotes a 47Ω resistor probability.

$$P(B) = \frac{62}{100}$$

P(B) denotes resistor with 5% tolerance

$$P(C) = \frac{32}{100}$$

<u>Joint prob</u>

$$P(A \cap B) = P(AB) = \frac{28}{100}$$

$$P(BC) = \frac{24}{100}$$

$$P(CA) = 0.$$

C & A are disjoint elements.

<u>conditional</u> <u>probability:</u>

$$P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{28}{100}}{\frac{62}{100}} = \frac{28}{62}$$

$$P\left(\frac{A}{C}\right) = \frac{P(A \cap C)}{P(C)} = \frac{0}{\frac{32}{100}} = 0$$

$$P\left(\frac{B}{A}\right) = \frac{P(B \cap A)}{P(A)} = \frac{28}{100} / \frac{44}{100} = \frac{28}{44}$$

$$P(B/C) = \frac{P(BC)}{P(C)} = \frac{\frac{24}{100}}{\frac{32}{100}} = \frac{24}{32}$$

$$P\left(\frac{C}{A}\right) = \frac{P(AC)}{P(A)} = \frac{0}{\frac{44}{100}} = 0.$$

$$P\left(\frac{C}{B}\right) = \frac{P(BC)}{P(B)} = \frac{24}{100} / \frac{62}{100} = \frac{24}{62}$$

$\frac{A}{B}$ $\frac{B}{C}$ $\frac{C}{A}$

$\frac{A}{B}$ $\frac{B}{C}$ $\frac{C}{A}$

10/10/22

Random Variable is a real value function.
Denotes with X(s).

1(i) where,
X denotes Random Variable
s denotes sample space.

Sample space (S) holds the sample points (s) (or) (s)

$$S = \{s_1, s_2, s_3, s_4 ----\}$$

Classification of Random Variable:

1. Continuous Random Variable
2. Discrete Random Variable
3. Mixed Random Variable

1. Continuous Random Variable: If any random variable holds the continuous range of variables then that random variable is called as continuos random variable.

$$Ex: \{-3 \leq X(s) \leq 10\}$$

2. Discrete Random Variable: The Random Variable which holds the discrete values then that random Variable is called as discrete Random Variable.

$$Ex: X(s) = \{-2, 0, 4, 8.5\}$$

3. Mixed Random Variable: If any random Variable holds the discrete as well as continous range of values then that random Variable is called as M.R.V.

$$Ex: X(s) = \{-4, 0, -2 \leq X(s) \leq 8\}$$

1(ii)

Probability Distribution function (PDF) / (CPDF) cummulative.

The probability of the event $\{X \leq x\}$ is called as PDF.
It is represented as $P\{X \leq x\}$ where, X - Random Variable.
x - Sample value

It is denoted with, $F_X(x) = P\{X \leq x\}$

It is also called as Cummulative probability Distribution function, shortly it is expressed as CPDF/PDF.

Properties of Distribution function ($F_X(x)$)

1. $F_X(-\infty) = 0$
2. $F_X(\infty) = 1$
3. $0 \leq F_X(x) \leq 1$
4. $F_X(x_1) \leq F_X(x_2)$ where, $x_1 \leq x_2$
5. $P\{x_1 < X \leq x_2\} = F_X(x_2) - F_X(x_1)$
6. $F_X(x^+) = F_X(x)$

Distribution function for the discrete random Variable:

If the random Variable is discrete then the distribution function can be expressed as $F_X(x)$

$$\boxed{F_X(x) = \sum_{i=1}^{N} P(x_i)\, u(x - x_i)}$$

where, $u(x) = \begin{cases} 1 & x \geq 0 \\ 0 & x < 0 \end{cases}$

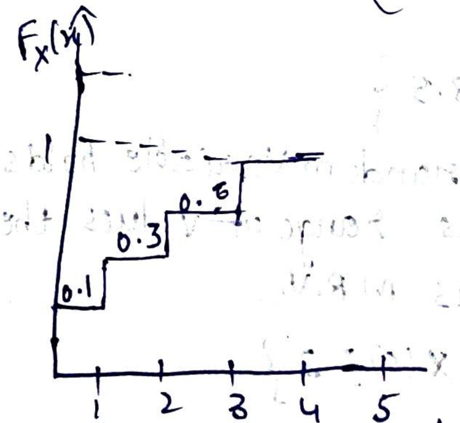


Fig : Probability Distribution function

Probability density function ($f_X(x)$) / Pdf:

Pdf is the derivative of the distribution function which can be expressed as,

$$\boxed{f_X(x) = \frac{d}{dx} F_X(x)}$$

where, $f_X(x)$ - Density function

$F_X(x)$ - distribution function

$\frac{d}{dx}$ - derivative.

probability density function can be shortly termed as "pdf".

Properties of pdf:

1. $0 \le f_X(x)$ - density must be always +ve.

2. $\int_{-\infty}^{\infty} f_X(x)\,dx = 1$.

3. $F_X(x) = \int_{x}^{\xi} f_X(\xi)\,d\xi$

4. $P\{x_1 < X \le x_2\} = \int_{x_1}^{x_2} f_X(x)\,dx$

PROBLEMS

1. A Random Variable X defining the values as following state whether X is continous (or) discrete (or) mixed Random Variable

a) $\{3 \le X(s) \le 8\}$

b) $\{-2, 0, 3, 5 \le X(s) \le 11\}$

c) $\{2, 4, 6, 2\}$

d) $\{9, 11, 18 \le X(s) \le 21\}$

a. Given that,

$X(s) = \{3 \le X(s) \le 8\}$

Here, Random variable has the continous range of values, then X is "continous Random Variable".

b. $X(s) = \{-2, 0, 3, 5 \le X(s) \le 11\}$.

Here, Random Variable X has discrete & continous values, then X is "mixed Random Variable".

c. $X = \{2, 4, 6, 2\}$

Here, R.V X has discrete values, then X is "discrete R.V".

d. $X(s) = \{9, 11, 18 \le X(s) \le 21\}$

Here, R.V X has discrete & continous values, then X is "Mixed R.V".

2. Random Variable X has the different instants $x_i = \{1,2,3,4,5,6\}$ & having the probabilities of $P(x_i) = \left\{\frac{2}{12}, \frac{1}{12}, \frac{4}{12}, \frac{3}{12}, \frac{1}{12}, \frac{1}{12}\right\}$ find the distribution at different instants and plot the distribution function.

A. Given,

$x_i = \{1,2,3,4,5,6\}$

corr. probabilities are, $P(x_i) = \left\{\frac{2}{12}, \frac{1}{12}, \frac{4}{12}, \frac{3}{12}, \frac{1}{12}, \frac{1}{12}\right\}$

Now,

Distribution function at individual instants is

$F_X(x) = P\{X \le x\}$

i.e for $x_i = 1$

$F_X(x) = P\{X \le x_1\} = \frac{2}{12}$.

for $x_i = 2$

$F_X(x) = P\{X \le x_2\} = \frac{2}{12} + \frac{1}{12} = \frac{3}{12}$

for $x_i = 3$

$F_X(x) = P\{X \le x_3\} = \frac{2}{12} + \frac{1}{12} + \frac{4}{12} = \frac{7}{12}$

for $x_i = 4$

$F_X(x) = P\{X \le x_4\} = \frac{2}{12} + \frac{1}{12} + \frac{4}{12} + \frac{3}{12} = \frac{10}{12}$

for $x_i = 5$

$F_X(x) = P\{X \le x_5\} = \frac{2}{12} + \frac{1}{12} + \frac{4}{12} + \frac{3}{12} + \frac{1}{12} = \frac{11}{12}$

for $x_i = 6$

$F_X(x) = P\{X \le x_6\} = \frac{2}{12} + \frac{1}{12} + \frac{4}{12} + \frac{3}{12} + \frac{1}{12} + \frac{1}{12} = 1$

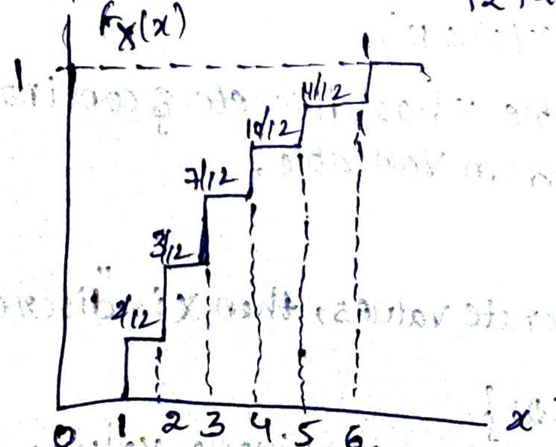


fig: plot of PDF

3. For the given random variable X probability density function defined as $f_X(x) = \begin{cases} Kx^2 & 0 < x \leq 2 \\ 0 & \text{otherwise.} \end{cases}$ then find Distributive function.

A. Given that, for the Random Variable X density function is,

$$f_X(x) = \begin{cases} Kx^2 & 0 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

Distribution function,

$$F_X(x) = \int_{-\infty}^{\infty} f_X(x)\,dx$$

$$= \int_{0}^{2} Kx^2\,dx$$

$$= K\frac{x^3}{3}\Big|_0^2 = K\frac{8}{3}$$

$$\boxed{\therefore F_X(x) = \frac{8K}{3}}$$

4. A Random Variable X has the density function $f_X(x) = \cancel{K}$

$$f_X(x) = \begin{cases} K \cdot \frac{x}{2} & -2 < x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

A. Given that, Random Variable X density function is,

$$f_X(x) = \begin{cases} K \cdot \frac{x}{2} & -2 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

Distribution function,

$$F_X(x) = \int_{-\infty}^{\infty} f_X(x)\,dx$$

$$= \int_{-2}^{2} K \cdot \frac{x}{2}\,dx$$

$$= \frac{K}{2}\,\frac{x^2}{2}\Big|_{-2}^{2} \;\; \cancel{= \frac{K}{2}\left(\frac{4+4}{2}\right) = \frac{8K}{4} = \frac{8K}{4}}$$

$$= \frac{K}{2}\left(\frac{4}{2} - \frac{4}{2}\right) = \frac{0K}{2} = 0K$$

$$\boxed{\therefore F_X(x) = 0K}$$

$$\boxed{\therefore F_X(x) = 0}$$

5. Random Variable x has the density function

$$f_X(x) = \begin{cases} Kx, & 0 < X \leq 1 \\ 0, & \text{otherwise} \end{cases}$$ find the constant K

A. Given that R.V x density f is,

$$f_X(x) = \begin{cases} KX, & 0 < X \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

To find the constant K the valid probability d.F = 1.

$$\int_{-\infty}^{\infty} f_X(x)\, dx = 1$$

$$\int_0^1 Kx \cdot dx = 1$$

$$K \int_0^1 \frac{x^2}{2} = 1$$

$$K \left[\frac{1}{2}\right] = 1$$

$$\frac{K}{2} = 1 \Rightarrow \boxed{K = 2}$$

$$\therefore f_X(x) = \begin{cases} 2x, & 0 < X \leq 1 \\ 0, & \text{o.w} \end{cases}$$

6. $$f_X(x) = \begin{cases} K(1-x), & 0 < X \leq 2 \\ 0, & \text{ow} \end{cases}$$

$$\int_{-\infty}^{\infty} f_X(x)\, dx = 1$$

$$\int_0^2 K(1-x)\, dx = 1$$

$$K \int_0^2 x - \frac{x^2}{2} = 1.$$

$$\cancel{K \left[\frac{-x^2}{2}\right]_0^2 = 1}$$

$$\cancel{K\left(\frac{-4}{2}\right) = 1}$$

$$\cancel{-2K = 1 \Rightarrow \boxed{K = \frac{-1}{2}}}$$

$$\cancel{\therefore f_X(x) = \{\tfrac{1}{2}}$$

$$K \int_0^2 x - \frac{x^2}{2} = 1$$

$$K[2-2] = 1$$

$$\boxed{K = 0}$$

Different density functions:

1. Binomial density function
2. poisson density function
3. Uniform density function
4. Exponential density-function
5. Rayleigh density function
6. Gaussian density function.

1. Binomial Random Variable is to define the Bernouli trails & game of having two chances & the radar and Sonar.

Now, the bernouli trail is nothing but game of two chances like winning (or) loosing. Here the binomial density function can be expressed as,

Impulse

$$\boxed{f_X(x) = \sum_{K=0}^{N} \binom{N}{K} p^N (1-p)^{N-K} \delta(x-K)}$$

Here, $\binom{N}{K}$ = Binomial coeffecient $= \frac{N!}{(N-K)!\,K!}$

Binomoil distrubution function can be written as,

$$\boxed{F_X(x) = \sum_{K=0}^{N} \binom{N}{K} p^N (1-p)^{N-K} u(x-K)}$$

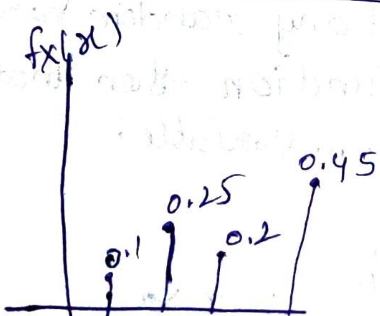


Fig. Binomial density fun.

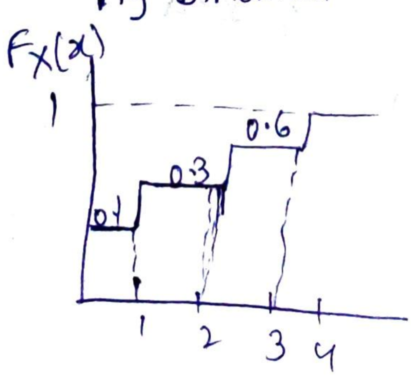


fig. Binomial distribution fun

2. poission density function: The poission density function for the random variable x is given by,

$$\boxed{f_X(x) = e^{-b} \sum_{K=0}^{\infty} \frac{b^K}{K!} \delta(x-K)}$$

where b = constant defining no. of events occuring during a particular time period.

As $K \to \infty, P \to 0$

Now, poission Distribution is

$$\boxed{F_X(x) = e^{-b} \sum_{K=0}^{\infty} \frac{b^K}{K!} u(x-K)}$$

3. Uniform density function: Uniform density function for the random variable x is given by

$$f_X(x) = \begin{cases} \frac{1}{b-a} & ; a \le x \le b \\ 0 & ; \text{otherwise} \end{cases}$$

where a & b are constants

$$F_X(x) = \begin{cases} 0 & ; x < a \\ \frac{x-a}{b-a} & ; a \le x < b \\ 1 & ; x \ge b. \end{cases}$$

4. Exponential density function: If any random variable X has the exponential density function then that random variable is, Exponentialle random variable.

It is expressed as

$$f_X(x) = \begin{cases} \frac{1}{b} e^{-(x-a)/b} & ; x > a \\ 0 & ; x < a \end{cases}$$

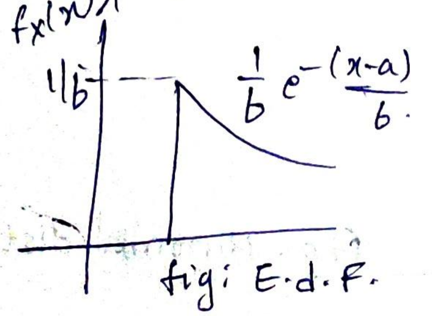
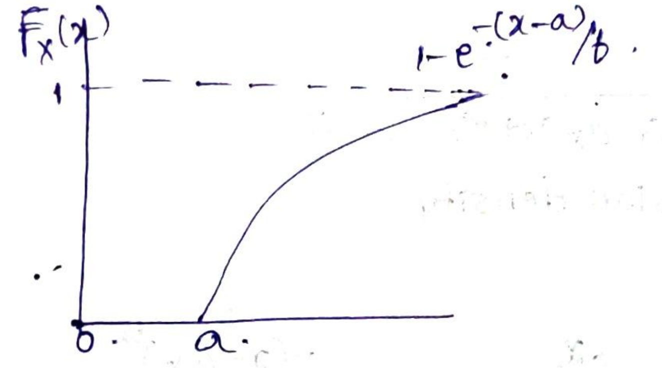


fig: E.d.F.

Distribution Function is,

$$F_X(x) = \begin{cases} 1 - e^{-(x-a)/b} & ; x > a \\ 0 & ; x < a \end{cases}$$

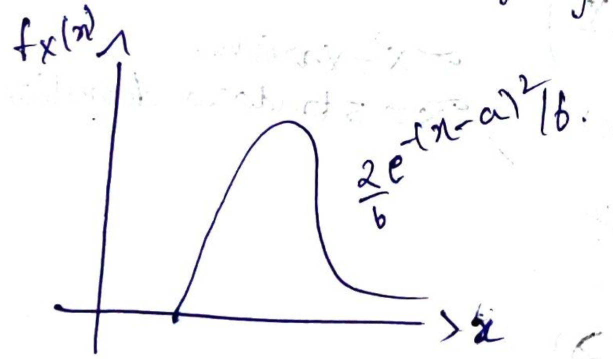


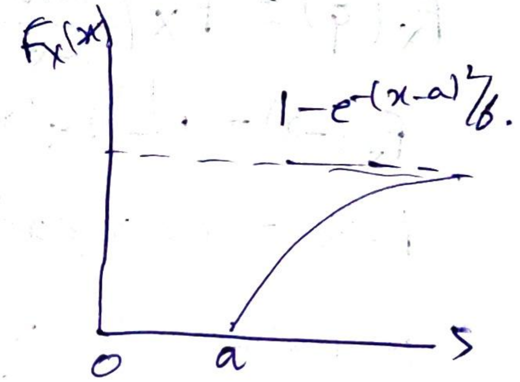
5. <u>Rayleigh density function</u>: If any r.v X has the Rayleigh density function then that random variable is called Rayleigh random variable

$$f_X(x) = \begin{cases} \frac{2}{b} e^{-(x-a)^2/b} & ; x > a \\ 0 & ; x < a \end{cases}$$

$$F_X(x) = \begin{cases} 1 - e^{-(x-a)^2/b} & ; x > a \\ 0 & ; x < a \end{cases}$$

x is called as Rayleigh R.V.





6. <u>Gaussian density function</u>: If any r.v 'X' has the gaussian density function then that random variable is called as Gaussian r.v.

$$f_X(x) = \frac{1}{\sqrt{2\pi \, \tilde{x}^2}} \, e^{\frac{-(x-a_x)^2}{2\tilde{x}^2}}$$

$\tilde{x}^2$ = Variance

a_x = Avg. value

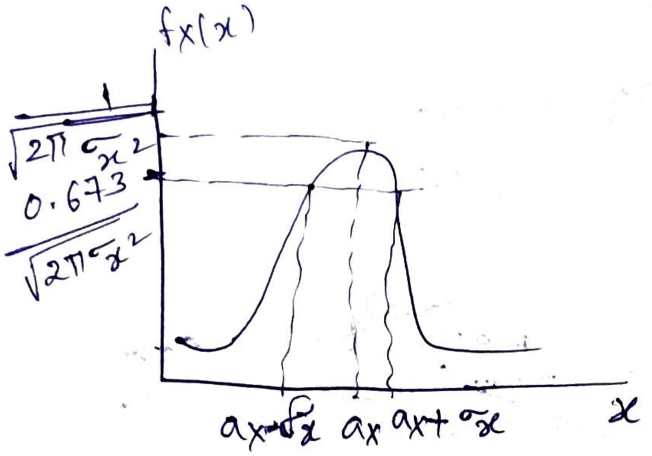


fig: Gaussian density

distribution

$$F_X(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi\sigma_x^2}} e^{\frac{-(x-a_x)^2}{2\sigma_x^2}}$$

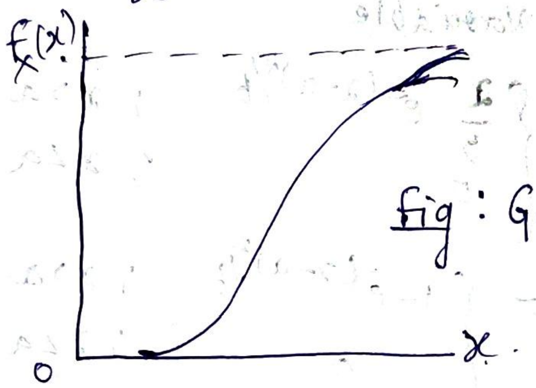


fig: Gaussian distribution.

$$F_X(\xi) = F_X\left(\frac{x-a_x}{\sigma_x}\right)$$

σ_x^2 = variance

σ_x = standard deviation.

$$F_X(-x) = 1 - F_X(x)$$

$$F_X(x) = P\{X \le x\}$$

$$1 - F_X(x) = P\{X > x\}$$

PROBLEMS

1. The probability of the raining by the cloud is 0.4 at an height 2Km. what is the probability to over come the rain by the eagle.

A. Given,

The raining probability of cloud is 0.4 at an height 2Km.

Then, the probability to overcome the rain by the eagle is

$$P(x > 2km) = 1 - P(x \leq 2km)$$

$$= 1 - 0.4 = 0.6.$$

∴ The probability to overcome the rain above the 2Km by the eagle is 0.6

2. For the random variable 'x', $x = 7.5$, $a_x = 2.5$, $\tilde{x} = 2$; then find $F_X(x)$.

A. Given,

R.V x, $x = 7.5, a_x = 2.5, \tilde{x} = 2$

then, $F_X(x) = F_X\left(\frac{x - a_x}{\tilde{x}}\right)$

$$= F_X\left(\frac{7.5 - 2.5}{2}\right)$$

$$= F\left(\frac{5}{2}\right) = F(2.5) = 0.9798.$$

Quie function

$$F_X(x) = 1 - Q(x)$$

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^{\alpha} e^{-\xi^2/c} \, d\xi.$$

$$Q(x) \simeq \left[\frac{1}{(1-a)x + a\sqrt{x^2 + b}}\right] \frac{e^{-x^2/2}}{\sqrt{2\pi}}$$

where,

$a = 0.339$

$b = 5.510.$

$F(2.5)$

$F(x) = 1 - Q(x)$

$F(2.5) = 1 - Q(2.5)$

$$Q(2.5) = \left[\frac{1}{(1-0.339)(2.5) + 0.339\sqrt{(2.5)^2 + 5.510}}\right] \frac{e^{-(2.5)^2/2}}{\sqrt{2\pi}}$$

$= 0.0202.$

$F_X(2.5) = 1 - Q(2.5)$

$= 1 - 0.0202 = 0.9798$

2. The Gaussian R.V $x = 3$ avg. value $= 0.5$ & standard deviation is 2 then find $F_X(x)$.

A. Given that, r.v x

$x = 3,\ a_x = 0.5,\ \sigma_x = 2$

$$F_X(x) = F\left(\frac{x - a_x}{\sigma_x}\right)$$

$$= F\left(\frac{3 - 0.5}{2}\right) = F\left(\frac{2.5}{2}\right) = F(1.25)$$

$F(1.25)$

WKT $F(x) = 1 - Q(1.25)$

$F(1.25) = 1 - Q(1.25)$

$$Q(1.25) = \left[\frac{1}{(1-0.339)1.25 + 0.339\sqrt{(1.25)^2 + 5.510}}\right] \frac{e^{-\frac{(1.25)^2}{2}}}{\sqrt{2\pi}}$$

$$= \left[\frac{1}{(0.661)1.25 + 0.339\sqrt{(1.5625) + 5.510}}\right] \frac{e^{-0.78125}}{\sqrt{2\pi}}$$

$= 0.10571$

$F_X(x) = 1 - Q(1.25)$

$= 1 - 0.10571$

$= 0.89429$

3. Assume that the height of clouds above the ground at some location is a Gaussian R.V x with $a_x = 1830m$ & $\sigma_x = 460m$. Find the prob that clouds will be higher then 2750m.

A. Given,

$a_x = 1830m$

$\sigma_x = 460m$

$P\{x > 2750\} = 1 - P\{x \le 2750\}$

$$F_X(2750) = F\left(\frac{2750 - 1830}{460}\right)$$

$$= F\left(\frac{920}{460}\right) = F(2)$$

WKT,

$F(x) = 1 - Q(x)$

$F(2) = 1 - Q(2)$

$$Q(2) = \left[\frac{1}{(0.661)2 + 0.339\sqrt{4 + 5.510}}\right] \frac{e^{-\frac{x^2}{2}}}{\sqrt{2\pi}}$$

$$= \left[\frac{1}{1.322 + 1.0176635}\right] \frac{e^{-2}}{\sqrt{2\pi}}$$

$= 0.023060562$

$P\{x > 2750\} = 1 - P\{x \le 2750\}$

$= x - (x - Q\{2750\})$

$= Q\{2750\} = 0.0228.$

4. Assume Automobile arrival at a gasoline station of poission & occur at an avg. rate of 50/hr the selection station has only one gasoline pump, if all cars are assumed to required 1min to obtain fuel., what is the probability that waiting line occur the pump.

A. Given,

$b = \lambda L$

$\frac{50}{hr}, \frac{50}{60} = \frac{5}{6}$, $P\{x>2\} = 1 - P\{x \le 1\}$

$F_X(x>2) = 1 - F_X(x \le 1$ $\left[F_X(x) = e^{-b} \sum_{K=0}^{x} \frac{b^K}{K} u(x-K) \right]$

$= 1 - F_X(1) - F_X(0)$

$= 1 - e^{-5/6} \sum_{K=0}^{n} \frac{(5/6)^K}{K!} u(x-K)$ $[\because 0! = 1]$

$= 1 - e^{-5/6} \left[\frac{(5/6)}{0!} \cdot 1 + \frac{(5/6)}{1!} \cdot 1 \right]$

$= 1 - e^{-5/6} [1 + 5/6] = 0.2033$

5. 10 coins are tossed 40 times what is the probability to get the 6 heads.

A. Given that,

No. of coins tossed $N = 10$

Probability to get the head is $\frac{1}{2}$

Here, the event is to get the 6 heads.

i.e, $K = 6$

Now, Binomial density can define the event as,

$= {}^{N}C_{K} (p)^{N-K} (1-p)^{N-K}$

$= {}^{10}C_{6} \left(\frac{1}{2}\right)^{10} \left(1 - \frac{1}{2}\right)^{10-6}$

$= {}^{10}C_6 \left(\frac{1}{2}\right)^{10}\left(\frac{1}{2}\right)^{4}$ $\quad\quad {}^{N}C_K = \frac{N!}{K!\,(N-K)!}$

$= {}^{10}C_6 \left(\frac{1}{2}\right)^{14}$

$= \frac{10!}{4! \times 6!}(0.00006) = \frac{10\times9\times8\times7\times\cancel{6!}}{4!\times\cancel{6!}} = 0.0126.$

$= 40 \times 0.0126$

$= 0.504.$

6. In a class there are 66 people what is the probability of attandance having 45 students for 60 days.

A. Given,

$N = 66$

$P = \frac{1}{2}$

$K = \cancel{60}.45$

Binomial density can define the event as,

$= {}^{N}C_K (P)^{N} (1-P)^{N-K}$

$\cancel{= {}^{66}C_{45} \left(\frac{1}{2}\right)^{66}\left(\frac{1}{2}\right)^{6}}$

$\cancel{= 90858768 \left(\frac{1}{2}\right)^{72}}$

$\cancel{= 8.906732 \times 10^{16} \left(\frac{1}{2}\right)^{72}}$

$= {}^{66}C_{45} \left(\frac{1}{2}\right)^{66}\left(\frac{1}{2}\right)^{21}$

$= \left(\frac{1}{2}\right)^{87} {}^{66}C_{45}$

$= 5.75584 \times 10^{-10}.$

$= 60 \times 5.75584 \times 10^{-10}$

$= 3.\cancel{5}45350 \times 10^{-8}$

Conditional probability : If A & B are any two events, the Occurance of A may depends on the Occurance of event B then that event is called as Conditional event,

Then that conditional event probability is called as, conditional probability.

$$P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0.$$

where, $P(A \cap B)$ = Joint probability

The Occurance of two events at a time commonly (jointly) then that event is called joint event.

joint event probability is called as joint probability.

Conditional Distribution :

$$F_X(x) = P\{X \leq x\}$$

$$F_X(x/B) = P\{X \leq x/B\}$$

The conditional distribution function can be expressed as, ↑, where x/B - conditional event.

$$P(X \leq x/B) = \frac{P(X \leq x \cap B)}{P(B)}; \quad P(B) \neq 0$$

$$= \int_{-\infty}^{\infty} f_X(x/B)\, dx$$

where, X is the event depends on the event B to Occur.

B is the proper event to define the $P(X/B)$

Properties of conditional distribution

1. $F_X(-\infty/B) = 0$
2. $F_X(\infty/B) = 1$
3. $0 \le F_X(x/B) \le 1$
4. $F_X(x_1/B) \le F_X(x_2/B)$; where, $(x_1/B) \le (x_2/B)$.
5. $P\{x_1/B < X/B \le x_2/B\} = F_X(x_2/B) - F_X(x_1/B)$
6. $F_X(x^+/B) = F_X(x/B)$

Conditional Density: The conditional density can be expressed as,

$$f_X(x) = \frac{d}{dx} F_X(x)$$

$$f_X(x/B) = \frac{d}{dx} F_X\left(\frac{x}{B}\right)$$

$\left(F_X\left(\frac{x}{B}\right) = \text{C.D.F} \quad f_X(x/B) = \text{c.d.f} \right)$

where, $\frac{d}{dx}$ = derivative of the conditional D.F

Properties of conditional density

1. $0 \le f_X(x/B)$
2. $\int_{-\infty}^{\infty} f_X(x/B)\,dx = 1$
3. $F_X(\xi/B) = \int_{x}^{\xi} f_X(\xi/B)\,d\xi$
4. $P\{x_1/B < X/B \le x_2/B\} = \int_{x_1}^{x_2} f_X(x/B)\,dx$.

PROBLEMS

1. 2 boxes Red, Green & Blue balls in them, the no. of balls of each colour is given in table. Our experiment will be to select a box & then a ball from the selected balls. One box is slightly larger than the other, causing it to be selected more frequently. Let B_2 be the event select the larger box, while B_1 is the event, select the smaller box. Assume $P(B_1) = \frac{2}{10}$, $P(B_2) = 8/10$. B_1 & B_2

are mutually exclusive events and $B_1 \cup B_2$ is the certain event.

Number of coloured balls in two boxes

x;	Ball colour	Box B_1	Box B_2	Total
1	Red	5	80	85
2	Green	35	60	95
3	Blue	60	10	70
Total		100	150	250

A. Given that,

Events are defining to choose the coloured balls from box 1 & box 2. The probability of,

$$Box\ 1 = P(B_1) = \frac{2}{10}$$

$$Box\ 2 = P(B_2) = \frac{8}{10}$$

choosing of coloured ball probabilities are

$$P(X=1/B_1) = \frac{5}{100} \quad ; \quad P(X=1/B_2) = \frac{80}{150}$$

$$P(X=2/B_1) = \frac{35}{100} \quad ; \quad P(X=2/B_2) = \frac{60}{150}$$

$$P(X=3/B_1) = \frac{60}{100} \quad ; \quad P(X=3/B_2) = 10/150 .$$

Now,

The conditional density function,

$$f_X(x/B_1) = \frac{5}{100}\delta(x-1) + \frac{35}{100}\delta(x-2) + \frac{60}{100}\delta(x-3)$$

$$f_X(x/B_2) = \frac{80}{150}\delta(x-1) + \frac{60}{150}\delta(x-2) + \frac{10}{150}\delta(x-3)$$

conditional distributives are,

$\cancel{F_X(x/B_1)}$ $F_X(x/B_1) = \frac{5}{100} u(x-1) + \frac{35}{100} u(x-2) + \frac{60}{100} u(x-3)$

$F_X(x/B_2) = \frac{80}{150} u(x-1) + \frac{60}{150} u(x-2) + \frac{10}{150} u(x-3).$

Now,

the total probability of choosing the coloured balls from box 1 & box 2

$P(X=1) = P(X=1/B_1) P(B_1) + P(X=1/B_2) P(B_2) = \frac{5}{100} \times \frac{2}{10} + \frac{80}{150} \times \frac{8}{10}$

$= 0.4366$

$P(X=2) = P(X=2/B_1) P(B_1) + P(X=2/B_2) P(B_2) = \frac{35}{100} \times \frac{2}{10} + \frac{60}{150} \times \frac{8}{10}.$

$= \frac{35}{500} + \frac{48}{150} = 0.39$

$P(X=3) = P(X=3/B_1) P(B_1) + P(X=3/B_2) P(B_2) = \frac{60}{100} \times \frac{2}{10} + \frac{10}{150} \times \frac{8}{10}.$

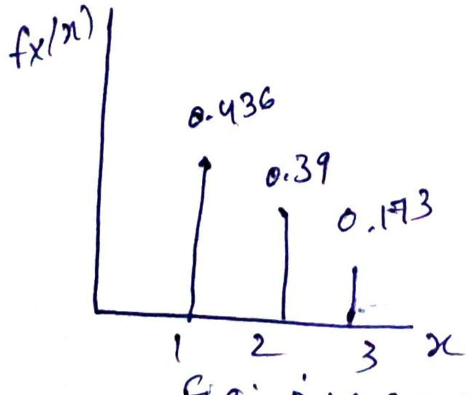
$= 0.12 + = 0.173$

The overall density function of choosing red, green & blue balls from the Box B_1 & B_2 is,

~~$f_X(x) = 0.4346 + 0.39 + 0.173$~~

$f_X(x) = 0.4366\,\delta(x-1) + 0.39\,\delta(x-2) + 0.173\,\delta(x-3)$

$F_X(x) = 0.4366\, u(x-1) + 0.39\, u(x-2) + 0.173\, u(x-3)$



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